

Feature-Funding Collapse on Scripted Data, and Why the Same Mechanism Explains Multi-Modal Policy Consensus

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The scripted-data failure: a precise causal chain

On the scripted data, the failure has a precise causal chain.

1. During the settle phase the actions are near-zero with essentially no local variation, and they are fully predictable from the end-effector height alone — so the loss demands no other feature there, and training prunes the local representation accordingly: the encoder’s local Jacobian participation ratio — the effective number of input directions the encoder still responds to — drops to ~ 1.5 –4 at settle versus 5–7 in the moving phases (from ~ 37 at initialization).
2. The labels *do* separate settle from the lift/transit stroke on-support (97–99% neighborhood purity, residuals at the noise floor), but this separation is only as wide as fitting requires: the standard MSE loss’s (L_2 ’s) separating gradient is residual-proportional, so it decays to zero once the training points are fit (measured: $+0.066$ early $\rightarrow 0.000$ at convergence), and it is built asymmetrically, since settle’s side pushes with near-zero residuals.
3. The stroke passes through the same height band, so for any state slightly off the demonstration support — beyond the thin separator — the height coordinate, the only direction with retained gain, assigns it to stroke content, and the policy reads out a full-magnitude translation during the zero-tolerance grasp window. This is confirmed at the behavioral level: the failure action matches the embedding-neighbor stroke labels ($\cos \approx +0.97$), and the quiet region’s basin boundary sits at only $\sim 40\%$ of the settle-to-stroke distance under L_2 .
4. Hetero-Gaussian (HG; heteroscedastic Gaussian NLL with a learned per-sample scale σ) fixes this at the formation stage: its $1/\sigma^2$ weighting gives the near-zero-residual distinctions $\sim 14\times$ the relative gradient share (versus $0.19\times$ under L_2), keeping the separating force alive after residuals shrink — the settle region retains $\sim 1.5\times$ more local feature directions, the basin boundary moves out to ~ 60 –70%, and the wrong-stroke readout stops being reachable in closed loop.

Relation to Multi-Modal Policy Consensus (arXiv:2509.23468)

The multi-modality paper plausibly faces the same issue, transposed from the phase axis to the modality axis. In contact-rich data, the tactile signal has exactly the properties our settle phase has: it is informative only during sparse contact windows (most of the time the touch channel is near-constant — the zero-local-variation condition), and on the clean training support it is almost entirely redundant given vision — the demonstrations can be fit to the noise floor from visual features alone, just as our settle actions can be regressed from the height axis

alone. So nothing in a monolithic training loss ever demands deep tactile features: gradient is allocated by explanatory share on the training distribution, vision explains nearly all of it, and the tactile directions of the fused representation are pruned the way our settle chart is pruned. The separation this leaves behind is again only as wide as fitting required — and deployment queries the unpaid-for region: under occlusion or perturbation, vision becomes ambiguous or misleading, and the policy must answer from a state where only the pruned tactile features could distinguish the right action. That is structurally identical to our drifted settle states being assigned by the height coordinate: a representation committed to the dominant channel, an off-support query, and a readout taken from whatever content the retained coordinate matches — which is why “vision overwhelms sparse but critical touch” is, in our terms, the funding-starvation collapse occurring along the modality axis.

Supporting experiments and ablations (scripted data)

Link 1 — pruning of the settle chart (Fig. 1)

The encoder’s local feature richness is measured as the participation ratio (PR) of the local Jacobian’s squared singular values — effectively, how many input directions the encoder still responds to at a given state. We compare *quiet* training states (small position-action magnitude, e.g. settle) against *loud* ones (large motions), for four losses: L_2 (plain MSE), HG and HT (heteroscedastic Gaussian and Student- t : a network head predicts a per-sample scale $\sigma(x)$, so the loss weights each sample by $1/\sigma^2$), and MIP (the two-view denoising flow-map objective our project studies). Training *prunes* all charts from a rich initialization; the losses differ in where the quiet window’s pruning floor lands. Causal controls: removing the height coordinate costs every model 0.36–0.72 of its function (all other groups ≤ 0.04); curated observations give the *same* collapse depth onto an unambiguous coordinate and a success rate (SR) of 100/100 ($\times 3$ seeds) — the disease requires the ambiguous survivor, not the collapse.

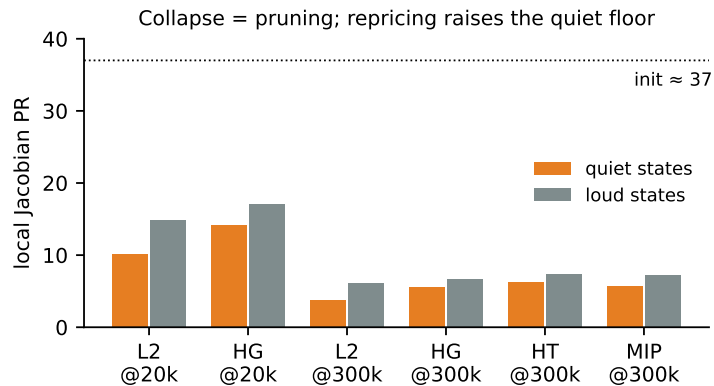


Figure 1: Quiet-vs-loud local PR across losses and training stages.

Link 2 — separation is only as wide as fitting requires (Table 1)

Intervention control: shortening the training chunk horizon to $H=2$ changes which states share targets; there, where the target geometry changes, the fold moves to the newly colliding pairs

	L_2	repriced / anchored
on-support neighborhood purity	97–99%	97–99%
teacher-forced residual (floor)	0.0016	0.0008
separating force, 39k $\rightarrow$ 300k	+0.066 $\rightarrow$ 0.000	MIP retains +0.15
merge cost, label-similar pairs	$\approx$ free (0.95)	13 $\times$
off-tube settle purity ($d \geq 2$)	3%	0–7% (universal)

Table 1: Supervision separates the training points for every loss; only the repriced losses (HG/HT) and the anchored one (MIP) keep paying for the margin after residuals reach the floor. “Off-tube” means normalized distance $d \geq 2$ from the nearest demonstration state, i.e. off the data support. The off-tube fold is universal — the failure site, not the cause.

(settle $\rightarrow$ approach 79%, shoulder $\rightarrow$ post 89%; healthy $H=10$ control) — the learned metric follows the labels by construction.

Link 3 — off-support readout and its execution dependence (Fig. 2)

The failure action equals the embedding-neighbor label mean ($\cos \approx +0.97$; zero alignment to observation-space neighbors), and canonicalizing the gripper coordinate collapses the cross-phase assignment (62% $\rightarrow$ 1%). Fig. 2 (left) shows the basin geometry; the right panel shows that the wrong readout costs success only when executed open-loop (per-cycle windows: L_2 corrective-head/expansive-tail $-0.17/ -0.17/+0.08$ vs. HG monotone contraction $-0.28/ -0.35/ -0.42$).

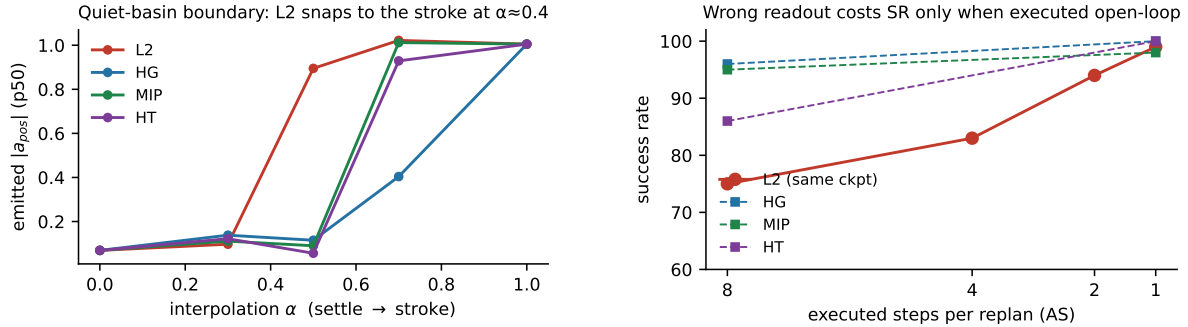


Figure 2: Left: emitted action magnitude along the settle $\rightarrow$ stroke interpolation. Right: success vs. executed steps per replan (same checkpoints).

Link 4 — the hetero-Gaussian mechanism and outcomes (Fig. 3, Tables 2, 3)

How each step of the explanation is established

Q2 first: does the representation actually collapse, and how is that measured?

Collapse is measured on the trained encoder, per state, by three quantities: the participation ratio (PR) of the local Jacobian’s squared singular values — the effective number of input directions the encoder responds to; the condition-number-style ratio $\kappa_{10} = s_1/s_{10}$ (how dominant the top direction is over the tenth); and the largest input column’s share of the Jacobian’s Frobenius

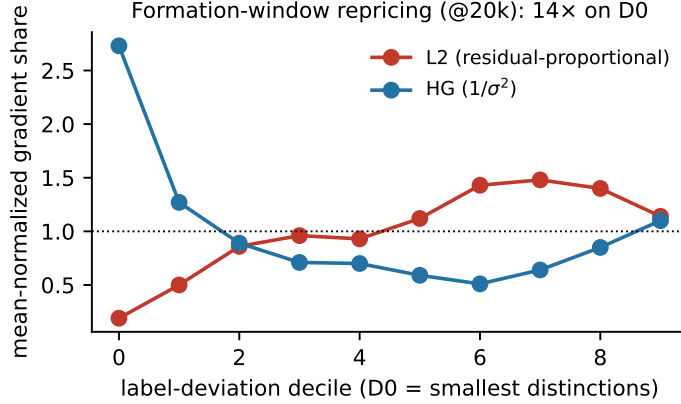


Figure 3: Gradient share by label-deviation decile at the 20k snapshot: $1/\sigma^2$ re-prices the smallest distinctions $14\times$ relative to L_2 and suppresses the outlier pocket; σ collapses to its floor by 300k (a formation-window channel).

3 training seeds	AS= 8 (standard)	AS= 1
L_2	{75, 66, 70}	{98, 75, 92}
hetero-Gaussian	{96, 97, 88}	{100, 95, 95}
L_2 3-seed ensemble	83	96

Table 2: Seed families are disjoint at the standard protocol: repricing acts as a *selector* (variance collapse), not only a mean shift. The checkpoint ensemble recovers about half the gap (posterior averaging).

mass. Table 4 collects the measurements. Reading it: the encoder starts isotropic ($\text{PR} \approx 37$); L_2 training prunes the *quiet* states’ chart far harder than the loud states’ (PR 3.8 vs. 6.1; κ_{10} 8.3 vs. 4.8 — and at the narrow settle band itself, PR 1.6 with $\kappa_{10} \approx 12.6$); the snapshots show this is monotone pruning of an initially rich chart, already under way at 20k; and the repriced losses end with both a higher quiet floor and a smaller quiet-vs-loud asymmetry. The surviving direction under L_2 is a height composite spread over ~ 12 redundant static-object columns (pairwise $|\cos| = 0.89$), which is why the top-column share stays low even as the chart collapses — the committed feature hides across many correlated columns rather than in one.

Q1: how do we know the collapse is caused by the claimed cause?

The claimed cause is the absence of local label variation (nothing demands that the output change with the state there) combined with L_2 ’s residual-proportional allocation. Three interventions isolate it:

- **Manipulate label variation, hold everything else fixed:** injecting 5 mm of label noise raises settle richness $1.9 \rightarrow 4.1$; human tremor (natural variation) gives 5.6–5.9, and the trained human encoder shows no quiet-vs-loud gap at all (last row of Table 4); the collapse tracks the variation dose and appears *only* where labels are locally constant (per-phase map matches the label-variation map; human datasets show no quiet-vs-loud gap).

intervention family	example (SR)	verdict
static reweighting	Cauchy 57; residual-w. 66; rel-err 72	fails
representation regularizers	contrastive 8; isotropy 2–19; isometry 86*	fails
smoothing / Jacobian penalties	obs-noise 68; jacreg 1–24	fails
adaptive repricing ($1/\sigma^2$)	hetero-Gaussian 96	works
label-variation injection	progaux (L_2 + episode-progress output) 94	works
observation curation	minimal (16-dim curated obs) 100	works
short-horizon ladder ($H=2$)	L_2 71 / MIP 81 / HG 91	ordered by repricing strength

Table 3: Specificity controls. *stressreg attains the flattest geometry measured yet leaves the fold intact — rank without content does not cure.

checkpoint	PR		κ_{10}		top-column share	
	quiet	loud	quiet	loud	quiet	loud
initialization	≈ 37 (isotropic)		—	—	—	—
L_2 @20k	10.1	14.9	3.6	2.8	.05	.04
L_2 @300k	3.8	6.1	8.3	4.8	.07	.05
settle band only	1.6	—	$\approx$ 12.6	—	—	—
HG @20k	14.2	17.0	2.9	2.6	.04	.03
HG @60k	8.3	10.5	4.1	3.4	.05	.04
HG @300k	5.5	6.7	5.7	4.5	.06	.05
HT @300k	6.3	7.3	5.1	4.3	.05	.05
MIP @300k	5.7	7.2	5.3	4.3	.07	.05
tool-hang <i>human</i> , L_2	16.4	18.1	2.5	2.3	.03	.03

Table 4: Collapse measurements: local Jacobian PR, κ_{10} , and top-column Frobenius share at quiet vs. loud training states. “Quiet” = position-action magnitude below the 40th percentile; “settle band” = the narrow grasp-alignment window only. The human-data row is the negative control: where quiet windows carry label variation, no quiet-vs-loud collapse gap forms.

- **Manipulate the allocation, hold the data fixed:** replacing L_2 by the $1/\sigma^2$ -weighted loss — whose funding shift we verify directly in the gradient tables ($14\times$ relative share on the smallest distinctions at 20k) — raises the quiet pruning floor $3.8 \rightarrow 5.5$ on identical data. Same data, different funding, different collapse depth.
- **Localization control:** the collapse sits exactly at the states whose labels are quiet, at every loss, in proportion to that loss’s residual funding (quiet/loud PR ratio 0.62 for L_2 vs. 0.79–0.86 repriced; Table 4).

Q3: how does the collapse influence success rate?

Not directly — and the factor-isolation experiments are the point. Collapse harms SR only in conjunction with two other factors: an *ambiguous* surviving coordinate, and *open-loop exposure* of the off-support region. Each factor is switched off separately while the others are held:

The mediation is quantified: at the policy’s own off-support states the neighborhood purity drops to 3% and the emitted action equals the neighbor-label mean (cos +0.97); success is then monotone in the escape rate across all five losses (escape fraction 0.29/0.19/0.11/0.07/0.06 $\rightarrow$ SR 75/86/94/95/96). Two cautions keep the claim honest: richness is a *footprint* of funding, not

collapse	ambiguous survivor	executed open-loop	SR
yes	yes	yes	71–75 (full-obs L_2 , AS= 8)
yes	no	yes	100 (curated obs; same collapse depth)
yes	yes	no	99 (same checkpoint, AS= 1)
reduced	yes	yes	94–96 (repriced loss)

Table 5: Factor isolation: switching off any single factor removes the SR damage. The collapse influences SR only through the chain collapse $\times$ ambiguity $\rightarrow$ off-support misassignment $\rightarrow$ wrong full-magnitude readout $\rightarrow$ executed open-loop.

itself sufficient (the flattest-geometry regularizer scores only 86 with the misassignment intact, and the richest-chart repriced loss is the weakest of the three winners), and the same conjunction predicts *absence* of harm correctly — on human data, where quiet windows carry label variation, no collapse forms and the failure modes are entirely different.

Known limitations. The $H=2$ ladder and some matrix cells are single training seed (evaluation-seed replicated only); baseline horizons are $H=10$; whether the hetero-Gaussian effect flows through the transient weighting path or the σ -head’s feature shaping is under test (a decoupled control is being evaluated).